Physiology of the Defecation Reflex

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GIT module

Third year medical students

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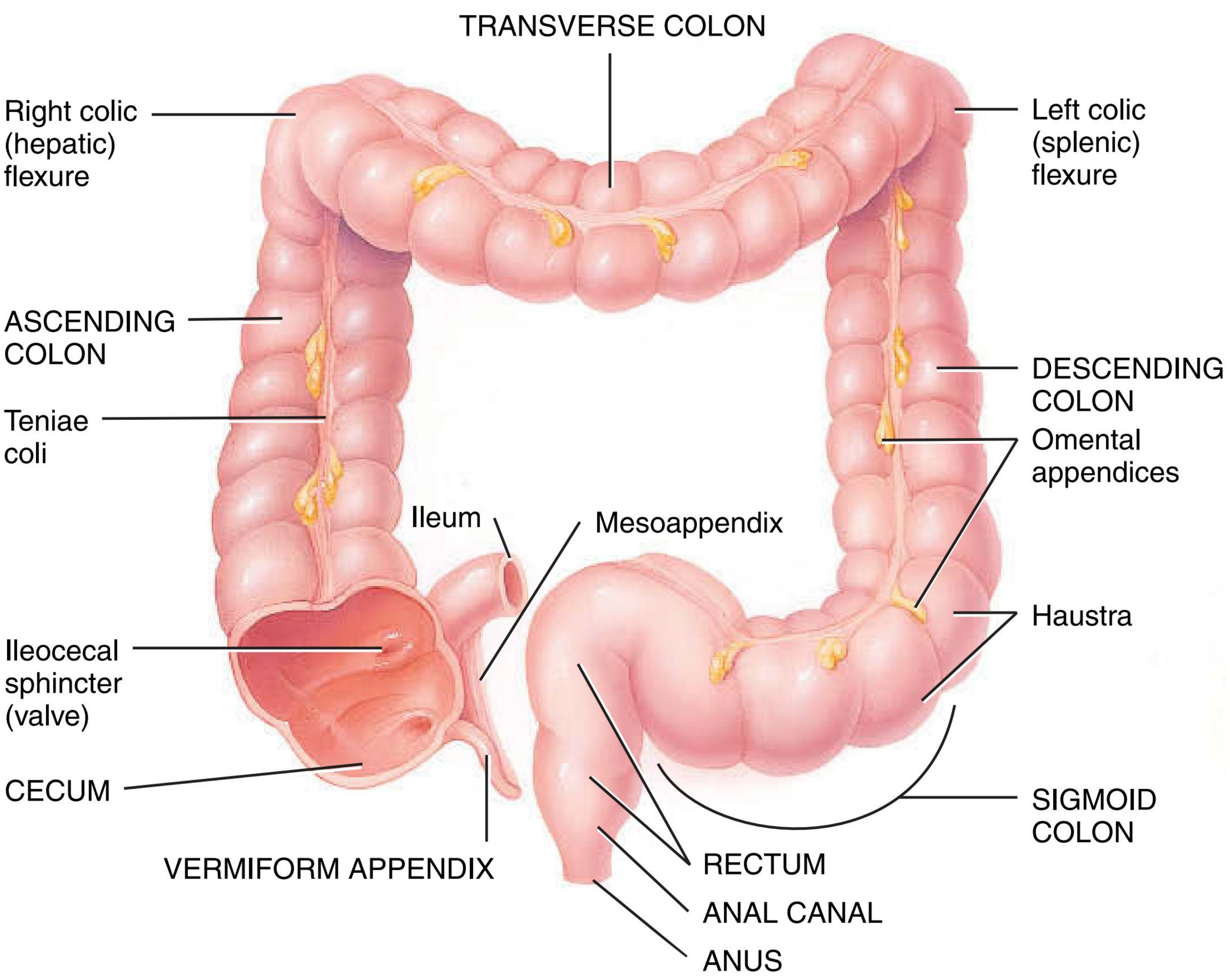
Objectives

By the end of this class, the students should be able to

- Understand the defecation reflex



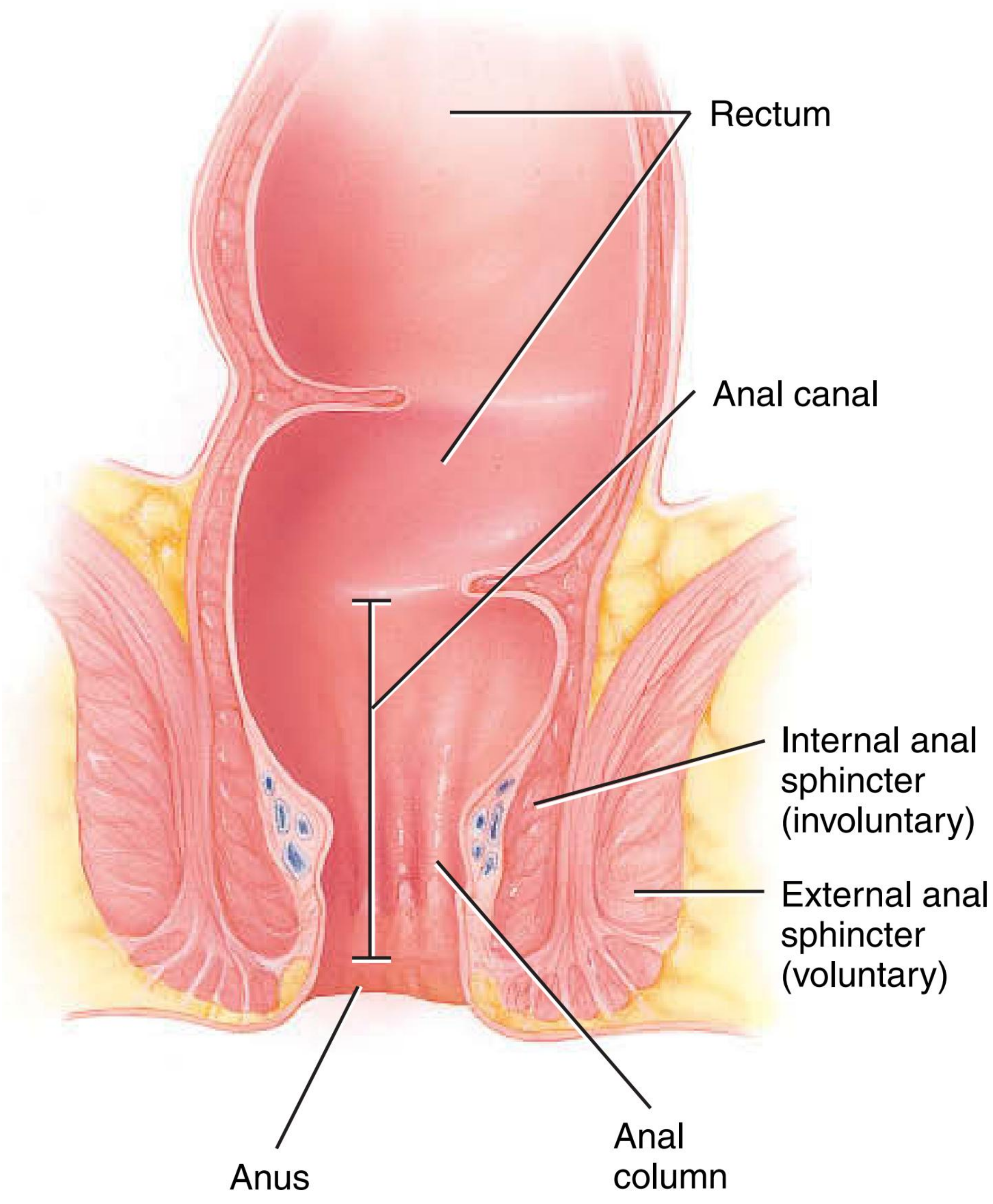
(a) Anterior view of large intestine showing major regions

FUNCTIONS OF THE LARGE INTESTINE

1. Haustral churning, peristalsis, and mass peristalsis drive contents of colon into rectum.
2. Bacteria in large intestine convert proteins to amino acids, break down amino acids, and produce some B vitamins and vitamin K.
3. Absorption of some water, ions, and vitamins.
4. Formation of feces.
5. Defecation (emptying rectum).

The Defecation Reflex

- Once the fecal material reaches the rectum
- The resulting *distension* of the rectal wall stimulates *stretch receptors*, which initiates a **defecation reflex** that results in defecation (the elimination of feces from the rectum through the anus)



(b) Frontal section of anal canal

The Defecation Reflex

<u>The defecation reflex occurs as follows:</u>

- In response to distension of the rectal wall, the receptors send sensory nerve impulses to the sacral spinal cord.
- Motor impulses from the cord travel along parasympathetic nerves back to the descending colon, sigmoid colon, rectum, and anus.
- The resulting contraction of the longitudinal rectal muscles shortens the rectum, thereby increasing the pressure within it.
- This pressure, along with voluntary contractions of the diaphragm and abdominal muscles, plus parasympathetic stimulation, **opens the internal anal sphincter**
- The **external anal sphincter is voluntarily controlled**.

The Defecation Reflex

- If the external anal sphincter is voluntarily relaxed, defecation occurs and the feces are expelled through the anus; if it is voluntarily constricted, defecation can be postponed.
- If defecation does not occur, the feces back up into the sigmoid colon until the next wave of mass peristalsis stimulates the stretch receptors, again creating the urge to defecate.
- In infants, the defecation reflex causes automatic emptying of the rectum because voluntary control of the external anal sphincter has not yet developed.
- The amount of bowel movements that a person has over a given period of time depends on various factors such as diet, health, and stress.
- The normal range of bowel activity varies from two or three bowel movements per day to three or four bowel movements per week

In summary- Key Mechanisms

- **Intrinsic reflex (enteric):** Rectal distension → local contraction of rectum + relaxation of internal anal sphincter.
- **Parasympathetic reflex (sacral cord S2–S4):** Amplifies rectal contraction and sphincter relaxation.
- **Voluntary control:** External anal sphincter + pelvic floor via pudendal nerve.
- **Central control:** Cortex decides to allow/withhold defecation → conscious relaxation of external sphincter & ↑ intra-abdominal pressure (Valsalva).